AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M269
Algorithms, Data Structures and Computability

Final Examination
Spring Semester 2024/2025

Date: 31/May/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

For each statement in list A, find the correct answer from list B. Please not the list B has more than 10 options from which to choose. Your answer should simply include the number from list A and its matching letter from list B. For example, like this [1 - Y], [2 – W] … and so on.

List A	List B
1) The bubble sort algorithm belong to … paradigm 2) The linear search searching algorithm is …….. than KMP algorithm 3) For an array of 128 sorted items, the *average* number of searches required to find an item using binary search is …….. 4) The number of comparisons involved in a linear search of an array A[n] is …….. 5) The ……. data structure is most suitable for representing the hierarchy in an army. 6) In Depth-First traversal, ……. nodes are processed first. 7) …….. is the best-case performance of looking up an element in a hash table (where n is the number of elements) 8) A tree is collection of line segments, called …….. and nodes called …….. that connected by the segments. 9) Searching for a value in a sorted list is the ……. case scenario 10) …….. is the recurrence relation that arises in relation to the complexity of binary search	a) Brute-force b) Worst c) 7 d) Greedy e) Faster f) T(n) = T(n/2) + k g) Slower h) 6 i) O(log n) j) O(1) k) Deepest l) Edges, vertices m) O(n) n) Tree o) T(n) = T(n-1) + k p) Graph q) Hash table r) T(n) = T(n/2) + O(n) s) Adjacent t) Average u) Vertices, edges v) Best w) Divide and Conquer x) T(n) = 2T(n/2) + k y) 5

PART 2: ALL QUESTIONS ARE REQUIRED [40 Marks]

Question 2: (20 marks)

a) What is the complexity of the following tasks? Give a rationale for your answer (10 marks: 1 mark for each, 0.5 mark is deducted if no rational is provided)

1) Serving the first person in a queue with n elements.
2) Find the most repeated element in a sorted array of size n.
3) Linear Search on a sorted sequence of length n.
4) Finding the kth order statistic in a list of n elements.
5) Incidence matrix representation of a graph with n vertices and m edges.
6) Insertion in a balanced Binary Search Tree of size n.

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7) Finding a sub-word of size m in a word of size using KMP algorithm.
8) Identifying the smallest values in a queue of size n.
9) Implementing BFS on a graph with ***n*** vertices and ***m*** edges.
10) Sorting a list of n identical items using Selection sort.

b) Determine a particular class of problems for each of the following descriptions: (total 4 marks: 2 marks each)
   1) Problems with clear yes-or-no answers are well-suited for computer analysis.
   2) While computers can quickly verify solutions, the process of finding those solutions can be time-consuming.

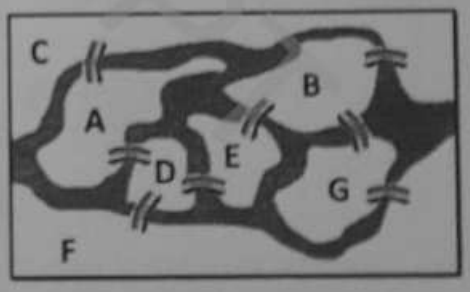
c) Give a short answer to the following questions: (6 marks)
   1) Given an array *arr* = [20, 21, 23, 25, 27, 30, 32, 35, 39] and *key* = 34, state in how many iterations would the linear search and binary search reach an answer? (2 marks)
   2) Demonstrate the insertion of [5, 28, 19, 15, 20, 33, 12, 17, 10] into a hash table with collisions resolved by chaining. Let the hash table have 7 slots, and let the hash function be h(k) = k mod 7. (4 marks)

Question 3: (20 marks)

a) Suppose that we have numbers between 1 and 100 in a binary search tree and want to search for the number 45. Which of the following sequences could be the right sequence of examined nodes? Justify your answer
   1) 5, 2, 1, 10, 39, 34, 77, 63, 45.
   2) 50, 25, 26, 27, 40, 44, 42, 45.
   3) 79, 50, 25, 26, 27, 40, 44, 45.
   [Hint: selecting the right sequence (2 marks), the justification (4 marks)]

b) Use your own simple words to describe each of the following paradigms, and give an example of the algorithms you know for each paradigm: (total 6 marks: 2 marks each)
   1) Divide-and-Conquer
   2) Brute force
   3) Greedy Algorithms

c) Consider the map in the illustrated figure. The map shows a group of separated lands connected by 8 bridges.
   1) Convert the map into a graph (2 mark)
   2) Why there is no Euler circuit in this graph? (2 marks)
   3) Add one bridge only to the above graph to create an Euler circuit and explain your answer. (2 marks)
   4) What is the order and the size of this graph?



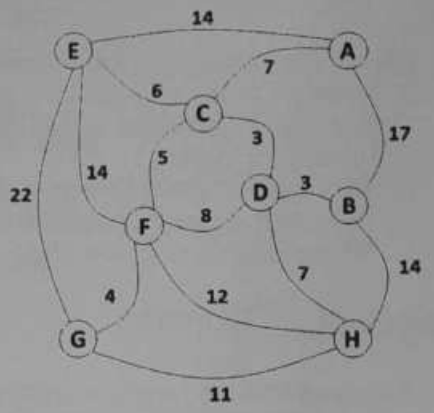
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PART 3: ALL QUESTIONS ARE REQUIRED [50 Marks]

Question 5: (40 marks)

Consider the weighted graph illustrated in Fig.1

a) Does this graph contain Euler circuit or/and Euler path? Why? (4 marks)
b) What is the sequence of visited node in BFS, and DFS if both algorithms start from node A? (4 marks)
c) Draw the minimum spanning tree of this graph. (6 marks). What is the name and complexity of such algorithm? (4 mark)
d) Use the Dijkstra's algorithm to determine the shortest paths from node (A) to all other cities (8 marks). Determine the shortest path and cost from node A to node H (2 marks).
[Hint: use the Dijkstra's table to implement the algorithm step by step]
e) Implement the proper algorithm to find the shortest paths between all pairs of nodes [Compute D_0, D_1, D_2]. Mention the name of the algorithm that could be used and its complexity (12 marks)



Question 6: (10 Marks)

You are given n points with integer coordinates X_1, X_2, X_3, ... and X_n on the X axis, where you can choose any of them as a base B. Let f(B) is the total distance from the chosen point B to all the rest n-1 of the points. Find the largest possible value of f(B) and the base B

[Note: The distance between points X_i and B is equal to $|X_i - B|$. For example, the distance between points X_1=5 and B = 8 is 3.]

Examples:

	Input	output	Explain
Test case 1:	[5, 1, 2, 7, 10]	f(B) =25, B =10	This value can be obtained if B is 10, This is because, f(10) = |10-5| + |10-1| + |10-2| + |10-7| + |10-10| = 25, if you choose any other base the f(b) will be smaller than 25.
Test case 2:	[11, 7, 5, 10, 2]	f(B) =25, B =2	This value can be obtained if B is 2, This is because, f(2) = |11-2| + |7-2| + |5-2| + |10-2| + |2-2| = 25, if you choose any other base the f(b) will be smaller than 25.

a) Design an algorithm to solve this problem with the least complexity (6 marks)
b) Derive the complexity of your algorithm. (4 marks)

End of questions

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